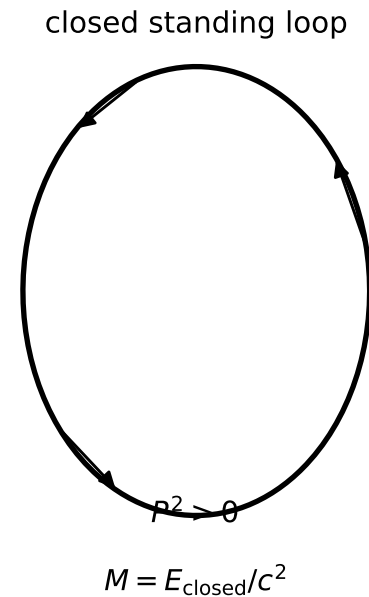
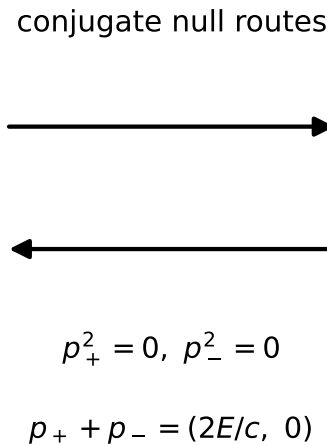
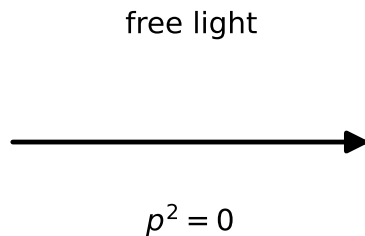


Null routes close into mass shell



Each route is individually lightlike. Closure cancels net spatial momentum while preserving energy, producing nonzero invariant mass.